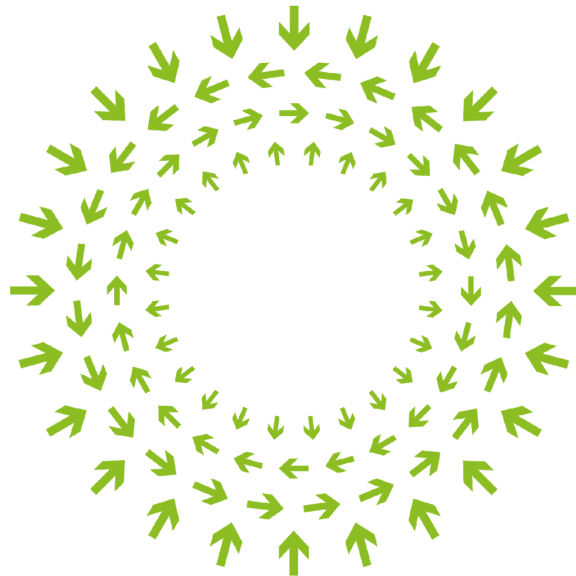




OPEN

Compute Project

Open Rack V3 HPR V2 12kW PSU

Rev: 1.0.0

06/12/2026

Authors:

Jingjing Sun
Mohammed Alisa
David Sun
Wenying Zhang
Dmitriy Shapiro
Mohammed Alisa
John Fernandes
Ben Kim
Jaswanth Vutukury
Baby Neil Escuadro
Siu Lam
Bo Wen
Zibo Chen

1. License

THE UPDATED DEFAULT CONTRIBUTOR LICENSE AGREEMENT (CLA) IS [OWFa 0.9](#). PLEASE VERIFY THE CORRECT CLA/FSA IS USED AND EXECUTED FOR THIS CONTRIBUTION.

1.1. Open Web Foundation (OWF) CLA

Contributions to this Specification are made under the terms and conditions set forth in **Modified Open Web Foundation Agreement 0.9 (OWFa 0.9)**. (As of October 16, 2024) (“Contribution License”) by:

META Platforms

Usage of this Specification is governed by the terms and conditions set forth in **Modified OWFa 0.9 Final Specification Agreement (FSA)** (As of October 16, 2024) (“**Specification License**”).

You can review the applicable Specification License(s) referenced above by the contributors to this Specification on the OCP website at <https://www.opencompute.org/contributions/templates-agreements>.

Notes:

The above license does not apply to the Appendix or Appendices. The information in the Appendix or Appendices is for reference only and non-normative in nature.

NOTWITHSTANDING THE FOREGOING LICENSES, THIS SPECIFICATION IS PROVIDED BY OCP "AS IS" AND OCP EXPRESSLY DISCLAIMS ANY WARRANTIES (EXPRESS, IMPLIED, OR OTHERWISE), INCLUDING IMPLIED WARRANTIES OF MERCHANTABILITY, NON-INFRINGEMENT, FITNESS FOR A PARTICULAR PURPOSE, OR TITLE, RELATED TO THE SPECIFICATION. NOTICE IS HEREBY GIVEN, THAT OTHER RIGHTS NOT GRANTED AS SET FORTH ABOVE, INCLUDING WITHOUT LIMITATION, RIGHTS OF THIRD PARTIES WHO DID NOT EXECUTE THE ABOVE LICENSES, MAY BE IMPLICATED BY THE IMPLEMENTATION OF OR COMPLIANCE WITH THIS SPECIFICATION. OCP IS NOT RESPONSIBLE FOR IDENTIFYING RIGHTS FOR WHICH A LICENSE MAY BE REQUIRED IN ORDER TO IMPLEMENT THIS SPECIFICATION. THE ENTIRE RISK AS TO IMPLEMENTING OR OTHERWISE USING THE SPECIFICATION IS ASSUMED BY YOU. IN NO EVENT WILL OCP BE LIABLE TO YOU FOR ANY MONETARY DAMAGES WITH RESPECT TO ANY CLAIMS RELATED TO, OR ARISING OUT OF YOUR USE OF THIS Open Compute Project • <3 Layer Specification Title> Date: XXXX, 2XXXX Page X SPECIFICATION, INCLUDING BUT NOT LIMITED TO ANY LIABILITY FOR LOST PROFITS OR ANY CONSEQUENTIAL, INCIDENTAL, INDIRECT, SPECIAL OR PUNITIVE DAMAGES OF ANY CHARACTER FROM ANY CAUSES OF ACTION OF ANY KIND WITH RESPECT TO THIS SPECIFICATION, WHETHER BASED ON BREACH OF CONTRACT, TORT (INCLUDING NEGLIGENCE), OR OTHERWISE, AND EVEN IF OCP HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

Table of Contents

1. License	3
1.1. Open Web Foundation (OWF) CLA	3
Table of Contents	4
Revision History	7
2. Scope	7
2.1. Glossary of Acronyms	8
3. Overview	10
4. Electrical Requirements	10
4.1. AC Input Voltage & Frequency	10
4.2. Start-up Sequence	11
4.3. Turn on – Cold Start	11
4.4. Turn on – AC Failure / Recovery	11
4.5. Input Voltage OVP/UVP Protection	12
4.6. Input Over-Current Protection	12
4.7. AC Inrush Current & Re-rush Current	13
4.8. Efficiency	13
4.9. Power Factor	14
4.10. Total Harmonic Current Distortion (THD)	14
4.11. DC Output voltage	14
4.12. DC Output set point voltage accuracy	14
4.13. DC Output droop voltage	14
4.14. DC Output Voltage Ripple & Noise	14
4.15. Dynamic Response	15
4.16. Voltage Overshoot/Undershoot at Turn on / Turn off	16
4.17. Hold-up Time	16
4.18. PSU-BBU transition with AC_Loss_L Signal	16
4.19. Forced discharge mode	17
4.20. Pulse load management mode	17
4.21. Pulse power requirements	17
4.22. Output Over Voltage Protection (OVP)	18
4.23. Over Power / Current protection	19
4.23.1. Over Power/Current Protection Limits	19
4.24. Output Short-circuit Protection	19
4.25. Over Temperature Protection	19
4.26. Active Current Sharing Accuracy through analog bus	19
4.27. PSU physical addressing	20
4.28. LED	20
4.29. Grounding	20
4.30. Y-Cap	21
4.31. Random timer	21
4.32. Transition and synchronization Requirements	22
4.32.1. Start-up (0V to 50V) Transition procedure	22

4.32.2. AC_Fault_CLR / Sync_Stop	22
4.33. Power system stability requirements	23
4.34. PSU connector pinout	23
4.35. Interoperability	26
5. Communication	26
5.1. Firmware Upgrades	26
5.2. Metering Accuracy	27
5.3. Blackbox Function	28
5.3.1. Fault Log History	29
6. Environment	29
6.1. Temperature	29
6.2. Humidity	29
6.3. Gaseous Contamination	29
6.4. Altitude	29
6.5. Acoustic Noise	29
6.6. Vibration and Shock (non-packaged)	29
6.7. Package Vibration, Drop and Compression	32
7. Thermal	32
7.1. Operational	32
7.1.1. Airflow Direction	32
7.1.2. DeltaT	32
7.1.3. Thermal Margin	33
7.2. Thermal design requirements	33
7.2.1. Sensor Accuracy:	33
7.2.2. System Fan	33
7.2.3. Surface Temperature	33
7.2.4. Rear Side Clearance	34
7.3. Fan Failure	34
7.4. PSU Thermal Monitoring	34
7.5. Shelf Output Busbar Clip Temperature Monitoring and Protection - if Busbar Clip is Selected	34
8. Mechanical	35
8.1. PSU Physical Dimensions	35
8.2. Construction	36
8.3. Materials and Fasteners	36
8.3.1. PCR Targets & Implementation	36
8.3.2. Supplier Qualification & Material Testing	37
8.3.3. Documentation & Certification	37
8.4. Latch & Handle Requirements	38
8.5. EMI Gaskets	38
8.6. Chassis Interface	39
8.7. Rear Blind Mate Connector	39
8.8. Rear Blind Mate Connector Protection	39
8.9. Mechanical Drawings	40

9. Reliability and Quality	41
9.1. Derating Design	41
9.2. MTBF Requirements	42
9.3. Design Failure Mode and Effect Analysis (DFMEA)	42
9.4. High Accelerated Life Test (HALT)	42
9.5. NPI Reliability Tests	42
9.6. Disallowed Components	43
9.6.1. Capacitors	43
9.6.2. Sheet Metal Material and Zinc Whisker Implications	43
9.6.3. Mix-source Builds	44
10. Manufacturing Quality Control and Testing	44
10.1. Burn-In (BI) and Ongoing Reliability Testing (ORT)	44
10.2. Hipot and Ground Continuity Test	45
10.3. DFT and DFM	45
10.4. Quality Control Process	45
10.5. Mass Production First Article Samples	45
10.6. Spec Compliance, Quality FA, Warranty	46
10.7. Conformal Coating	46
11. Compliance requirements	46
11.1. Safety Standards	47
11.1.1. Safety requirements	47
11.2. EMC Requirements	47
11.3. Environmental Compliance	48
11.4. Documentation	49
12. Compliance with OCP Tenets	50
12.1. Openness	50
12.2. Efficiency	50
12.3. Impact	50
12.4. Scale	50
12.5. Sustainability	50

Revision History

Date	Version	Change By	Comments
6/12/2026	1.0.0	Jingjing Sun David Sun Mohammed Alisa Wenying Zhang Ben Kim Baby Neil Escudro Siu Lam	Initial OCP Release

2. Scope

This document defines the technical specifications for Open Rack V3 high power rack (HPR) V2 12kW Power Supply Unit (PSU) used in the Open Compute Project.

2.1. Glossary of Acronyms

The following acronyms and abbreviations are used throughout this specification:

Acronym	Definition
AC	Alternating Current
ASTM	American Society for Testing and Materials
AVL	Approved Vendor List
BBU	Battery Backup Unit
BI	Burn-In
BOM	Bill of Materials
CAD	Computer-Aided Design
CB	Certification Body
CE	Conformité Européenne (European Conformity)
CFM	Cubic Feet per Minute
CISPR	Comité International Spécial des Perturbations Radioélectriques (International Special Committee on Radio Interference)
CLA	Contributor License Agreement
COG	Ceramic of Glass (C0G capacitor dielectric)
CPK	Process Capability Index
CSA	Canadian Standards Association
DC	Direct Current
DFM	Design for Manufacturability
DFMEA	Design Failure Mode and Effect Analysis
DMTBF	Demonstrated Mean Time Between Failures
DVT	Design Verification Test
ECHA	European Chemicals Agency
EEE	Electrical and Electronic Equipment
EIA	Electronic Industries Alliance
EMC	Electromagnetic Compatibility
EMI	Electromagnetic Interference
EPA	Environmental Protection Agency
EU	European Union
EVT	Engineering Verification Test
FA	Failure Analysis
FAI	First Article Inspection
FCC	Federal Communications Commission
FET	Field-Effect Transistor
FW	Firmware
GPU	Graphics Processing Unit
GR&R	Gage Repeatability and Reproducibility
HALT	High Accelerated Life Test
HVDC	High Voltage Direct Current
HW	Hardware
I2C	Inter-Integrated Circuit
IEC	International Electrotechnical Commission
IPC	Institute for Printed Circuits (now Association Connecting Electronics Industries)

ISTA	International Safe Transit Association
LED	Light-Emitting Diode
MOS/MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor
MTBF	Mean Time Between Failures
NPI	New Product Introduction
NPO	Negative-Positive-Zero (capacitor temperature coefficient)
NRTL	Nationally Recognized Testing Laboratory
OCP	Over-Current Protection / Open Compute Project (context-dependent)
ORT	Ongoing Reliability Testing
OTP	Over-Temperature Protection
OVP	Over-Voltage Protection
OWFa	Open Web Foundation Agreement
PC/ABS	Polycarbonate/Acrylonitrile Butadiene Styrene
PCBA	Printed Circuit Board Assembly
PCN	Product Change Notification
PCR	Post-Consumer-Recycled
PDU	Power Distribution Unit
PE	Protective Earth
PFAS	Per- and Polyfluoroalkyl Substances
PFMEA	Process Failure Mode and Effect Analysis
PMI	Power Management Interface
PMM	Power Management Module
PN	Part Number
PSU	Power Supply Unit
PVT	Production Verification Test
PWM	Pulse Width Modulation
RCCA	Root Cause Corrective Action
REACH	Registration, Evaluation, Authorisation and Restriction of Chemicals
RH	Relative Humidity
RMA	Return Merchandise Authorization
RoHS	Restriction of Hazardous Substances
ROM	Read-Only Memory
RPM	Revolutions Per Minute
SCP	Short-Circuit Protection
SEC	Securities and Exchange Commission
SELV	Safety Extra-Low Voltage
SGCC	Steel Galvanized Cold Commercial
SMT	Surface-Mount Technology
SN	Serial Number
SPC	Statistical Process Control
SW	Software
TOR	Top of Rack
UL	Underwriters Laboratories
UVP	Under-Voltage Protection
WCSA	Worst Case Stress Analysis
WEEE	Waste Electrical and Electronic Equipment

3. Overview

This spec defines the single phase 48V PSUs that fit into the 48V power shelf. The PSU is intended for use on a power shelf that is part of the rack, for supplying DC power to system loads. Several PSUs with a minimum of N+1 redundancy shall be included in the power shelf.

The PSU operates based on “narrow range 48V Architecture.” Based on this concept, the PSU output voltage is 50V with an adjustable range. Benefits of this architecture compared to Orv2 with 40~60V dc voltage range are:

- Voltage range as low as possible to eliminate oversized voltage/current design
- Widely enable 4:1 fixed ratio converters and downstream conventional 12V PoL converters.
- Power to flow naturally, not with Software control and dependency.
- Enable simple fixed-voltage PSU design.

4. Electrical Requirements

Requirements Brief Summary is given below:

- 12kW output power
- Input rated nominal voltage 200V to 277Vac with +/- 10%
- Peak efficiency > 97.5%, measured with fans
- Active power factor correction (meets EN/IEC 61000-3-2 and EN 60555-2 requirements)
- DC Output overvoltage and overcurrent protection
- AC Input overvoltage and undervoltage protection
- Over-temperature warning and protection
- Active current sharing on top of droop
- Hot insertion/removal (hot plug)
- Front to back air cooling
- Internally controlled variable-speed fan
- Ability to field FW upgrade (with bootloader)

4.1. AC Input Voltage & Frequency

The PSU shall be capable of supplying output power over single phase input voltage range of 180 – 305V, and frequency of 47 – 63 Hz. Table 1 specifies the AC input voltage and frequency requirements for continuous operation.

Parameter	Nominal	Min	Max	Output Power Rating (kW)
AC Input Voltage	200Vac-277Vac Typical at 220, 230, 240, 277Vac	180Vac	305Vac	12kW at 218Vac - 305Vac Output power derated to 10kW at 180 - 218Vac
Frequency	50Hz, 60Hz	47Hz	63Hz	

Table 1: AC Input Voltage & Frequency Requirements.

The table above specifies the AC input voltage and frequency requirements for continuous operation. PSU minimum and maximum operating voltage can be limited further by software. If the PSU input voltage and frequency go out of the range while operating at any load value, PSU shall power off its power factor correction (PFC) circuit and relay within 20ms.

4.2. Start-up Sequence

The PSU shall be able to start up under rated nominal power at the min AC input voltage as specified in Table 1 above. With AC present, within specified parameters, the PSU must always remain operational. The startup sequence shall be designed such that the PSUs are able to meet the overall system start-up time and inrush current requirements specified later.

After AC voltage is applied, the internal bias supply starts, the microcontroller boots and keeps the PFC off as well as the dc-dc converter. Then the microcontroller closes the inrush relay and turns on all the converters.

4.3. Turn on – Cold Start

The output voltage of the PSU shall be monotonic during turn on and turn off, no reverse voltage shall occur during turn off.

- The dc output voltage rise time without external capacitors should be less than **200ms** with 120% constant-current limit. If after 10s the output voltage is out of range, the PSU shall consider it as OCP and restart once.
- The max capacitive loading on a single unit at power up is 40 mF.
- For any loads (from 'no-load' to 'max-load'), the output voltage will rise monotonically from $0V_{DC}$ to $50V_{DC}$, without overshoot or ringing, at any turn on following application of AC input voltage, and anytime when the PSU resumes functionalities after an automatic protection condition (including parallel operations). The output voltage will fall monotonically from $50V_{DC}$ to $0V_{DC}$, without undershoot or ringing, at any AC loss, and at any turn off caused by an automatic protection condition (including parallel operations).
- Output voltage shall never reverse polarity at the turn off (all conditions and converters).

4.4. Turn on – AC Failure / Recovery

The PSU shall recover automatically after an AC power failure. The start-up time requirement shall be the same as that of a cold start specified in section above.

The PSU shall include a soft-start that promptly resets at any input AC loss > 20ms, or after any automatic protection conditions.

4.5. Input Voltage OVP/UV Protection

The PSU shall shut down if the input voltage is over 346Vac line-to-neutral (600Vac line-to-line) for 20ms.

The minimum and maximum AC input voltage and frequency protection shall allow the PSU to operate normally within the voltage/frequency ranges and line surge/sag condition in section 4.1. The voltage/frequency protection thresholds shall have hysteresis.

The PSU shall meet the voltage surge/swell requirements in ITIC.

The PSU with BBU shall be able to ride through any line sag/dip conditions (i.e. no interruption on output) within the BBU backup time, including IEC61000-4-34.

4.6. Input Over-Current Protection

The PSU shall incorporate primary fusing on both phase and return lines for input over-current protection to meet product safety requirements. Fuses shall be selected to prevent nuisance trips. Fuse may be internal to the unit and need not be user serviceable. AC inrush current shall not cause the fuse to blow under any conditions. No PSU operating condition shall cause the fuse to blow unless a component in the PSU has failed. This includes DC output overload and short-circuit conditions. Fuse shall be approved by UL for an interrupt rating of at least 40kA.

PSU fuse shall be in coordination with data center tap-off box fuse curve as given below. Two AC feeds go to the power shelf, one PSU per input AC phase feed.

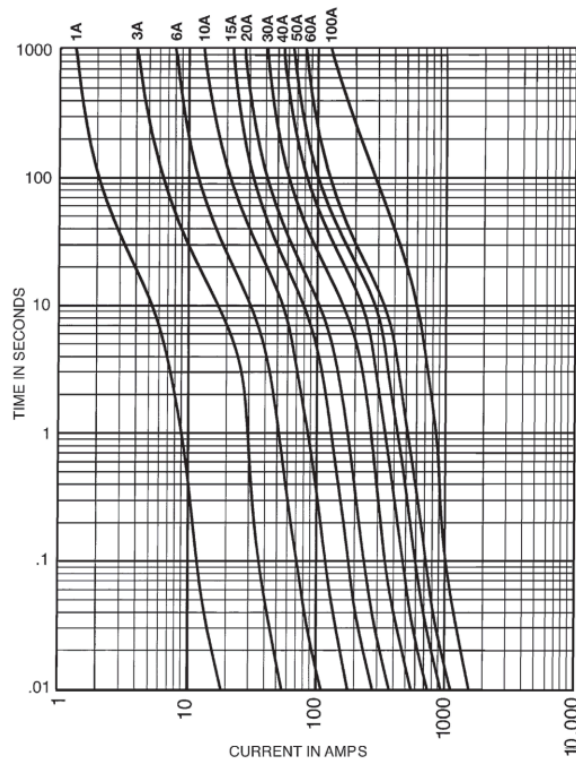


Figure 1: Data Center Tap-off Box Fuse Curve

NEC tap-off box is 60A rated, derating to 48A.

IEC tap-off box is 60A rated, no derating.

4.7. AC Inrush Current & Re-rush Current

1) AC Inrush Current

Maximum AC inrush current from cold power-on shall be limited to no greater than 60A RMS (85A peak) at any AC operating voltage. This specified inrush current shall not include the X-Capacitors charging.

2) Re-rush Current

Maximum AC re-rush current at AC recovery after dropout shall be less than five times of the steady-state current for less than 2ms. The steady-state current varies with the input AC voltage and loading levels. Some re-rush current limits are defined in Table 2 as an example for reference.

AC Voltage, RMS	Steady-state rated AC current, RMS	Re-rush Current Limit at 100% load, RMS	Re-rush Current Limit at 50% load, RMS
305 Vac	42A	5 x 42A	5 x 42A x 50%
291 Vac	44A	5 x 44A	5 x 44A x 50%
277 Vac	46.3A	5 x 46.3A	5 x 46.3A x 50%
230 Vac	55.7A	5 x 55.7A	5 x 55.7A x 50%
218 Vac	58.8A	5 x 58.8A	5 x 58.8A x 50%

Table 2: Re-Rush Current Limit Examples

This specified re-rush current shall not include the X-Capacitors charging.

4.8. Efficiency

The efficiency of the PSU when measured at an AC input voltage of 230V~277V and with the cooling fans connected (with input and output voltage measured at corresponding connectors, at 25°C ambient and after 30 minutes running at full load) shall meet the requirements outlined in Table below.

Load Range (%)	Peak Efficiency (%)	Min Efficiency (%)
30% to 100% of full load	> 97.5% @230, 240, 277V	> 96.5% @230, 240, 277V
10% to 30% of full load	-	94% @230V~277V

Table 3: Efficiency Requirements

4.9. Power Factor

The PSU shall incorporate an active power factor correction circuit such that the power factor meets the requirements at AC nominal voltage of 230V~277Vac as specified in Table 4.

Output load (% of rated output load)	PF
<10%	N/A
10% - 20%	>0.95
20% - 80%	>0.98
80% - 100%	>0.99

Table 4: Power Factor Requirements

The PSU input transient response should not cause an up-stream power quality issue.

4.10. Total Harmonic Current Distortion (THD)

For nominal input voltage of 230~277Vac:

Output load (% of max output load per module)	Maximum ITHD (%)
5-10	15
10-30	10
30-100	5

Table 5: Total Harmonic Current Distortion (THD)

4.11. DC Output voltage

The DC output voltage of the PSU shall be fixed at 50V @ NL and 49V @ FL while in normal operation. The PSU output voltage set point shall be adjusted via Modbus by +/-1V. The PSU shall supply rated power for the entire range of DC output voltage.

4.12. DC Output set point voltage accuracy

The PSU set point output voltage accuracy at 100% load shall be +/- 0.25% or +/- 0.125V for 49V. This can be accomplished by EoL calibration.

4.13. DC Output droop voltage

The PSU droop voltage (0%-100%) shall be 1V by default (with the tolerance of +/- 0.125V). Meaning that the output voltage is 49V at 100% load and 50V at no-load. The droop extends linearly to 150% (it means at 150%, the droop is 1.5V, and voltage is 48.5V).

4.14. DC Output Voltage Ripple & Noise

The DC output voltage ripple and noise shall not exceed 500 mV peak to peak. Ripple and noise are defined as periodic or random signals over a frequency band of 5Hz to 100MHz measured across a steady-state resistive load. Measurements shall be made differentially using an oscilloscope with 100MHz

bandwidth limit enabled. Compliance will be verified using a 0.1uF capacitor connected locally to the oscilloscope probe tips during this measurement.

4.15. Dynamic Response

Under these testing conditions, the DC output voltage shall not vary by more than specified for undershoot and overshoot with 3ms settling time (with and without capacitive loading on a single unit of 40mF). The measurement shall be from the new steady state value (due to droop) to the voltage spike/dip peaks during a transient load.

BBU backup should not be triggered during the dynamic loading operation.
The PSU input current waveform shall not be clamped during pulse loading operation.

Step load increase or decrease	Frequency	Transient load rate	Max undershoot and overshoot
50% step load (10% min. load)	Up to 10kHz	3 A/uSec	0.5V
90% step load (10% min. load)	Up to 6.5kHz	3 A/uSec	1V
140% step load (10% min. load)	Up to 4kHz	3 A/uSec	1.5V

Table 6: Dynamic Load Requirements

Dynamic load profiles under test include but not limited to the ones shown in Table 7.

No.	Low load	High load	Duty cycle	Frequency
1	10%	60%	50%	20Hz, 100Hz, 500Hz, 1kHz
2	50%	100%	50%	
3	10%	100%	50%	
4	10%	150%	50%	
5	10%	60%	50%	10kHz
6	50%	100%	50%	10kHz
7	10%	100%	50%	6.5kHz
8	10%	150%	50%	4kHz

Table 7: Example Dynamic Load Profiles

4.16. Voltage Overshoot/Undershoot at Turn on / Turn off

The output voltage overshoot/undershoot upon the application or removal of AC input voltage, under the specified input voltage defined before, shall not exceed 1V.

4.17. Hold-up Time

Minimum holdup time requirements at different load power are given in Table 8. Holdup time is calculated from the time AC input is lost till the time the output voltage is out of acceptable 48V range.

Load power	Holdup time (ms)
100%	20.00
110%	18.18
120%	16.67
130%	15.38
140%	14.29
150%	13.33

Table 8: Minimum Hold-up Time Requirement

4.18. PSU-BBU transition with AC_Loss_L Signal

AC_Loss_L signals have been added to the PSU and BBU (Battery Backup Unit) to enable control during AC loss events. Each PSU now has an output AC_Loss_L signal, while each BBU has two input AC_Loss_L signals.

The following is the diagram of AC_loss_L signal connection.

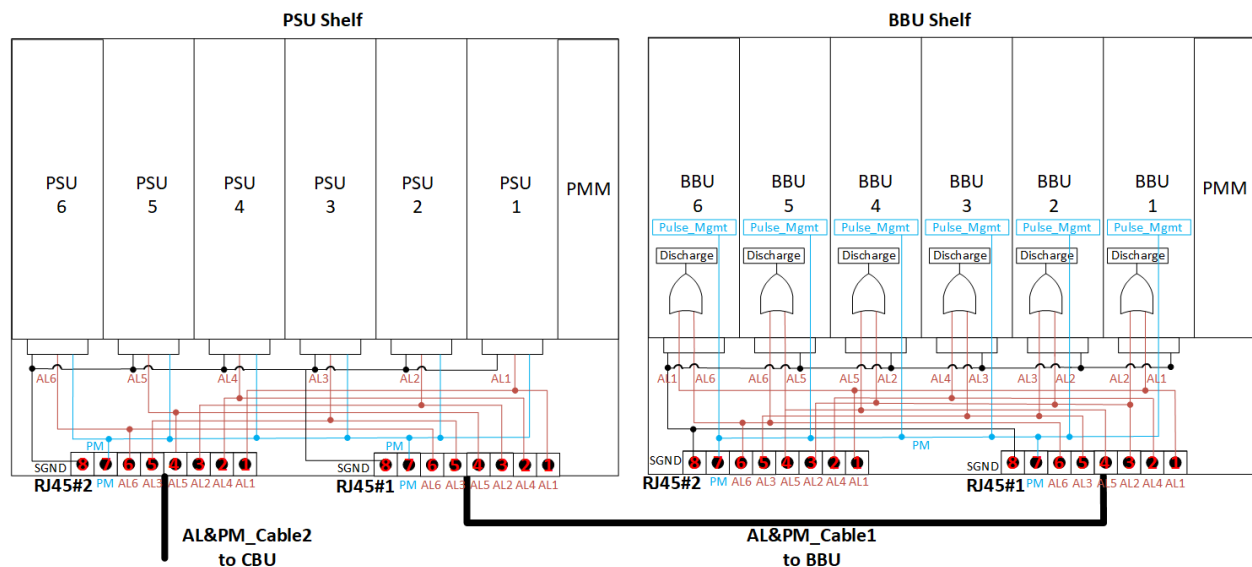


Figure 2: HPR V2 PSU/BBU AC_LOSS & PULSE_MGMT Signals for AC to BBU Transition

The AC_Loss_L signal assertion shall be triggered by the following conditions using a logic "OR" operation to ensure proper backup operation under all conditions.

- 1) Upon detection of AC loss, the PSU should assert the AC_Loss_L signal after a validation delay of 4-5ms.
- 2) The second condition for AC_Loss_L assertion will be based on the threshold of the remaining bulk capacitor energy, which should be kept at a minimum of > 5ms to allow sufficient time for the BBU to ramp up.
- 3) When other AC faults happen (input UVP/OVP/UFP/OFP), the PSUs should assert the AC_Loss_L signal after certain validation delays.

When a BBU receives the active AC_Loss_L signals, it will pull the Sync_Start_L signal low, allowing the BBU shelf to initiate immediate discharge.

After AC restores to normal condition, the PSU shall deassert the AC_Loss_L signal after a minimum of 500ms validation delay upon detecting the PSU main output is powered up. This ensures the safe transition from PSU to BBU and keeps the 50V bus voltage high.

The PSU AC_Loss_L signal can be toggled through Modbus for testing purposes, including cable connections, to ensure the proper battery backup operation.

To avoid BBU over-discharging due to single fault conditions like PSU input fuse open or PFC faults, the PSU shall deassert the AC_Loss_L signal after AC_Loss_L asserted for 300s.

4.19. Forced discharge mode

The rack monitor has the capability to send Modbus commands to PSUs in order to toggle the AC_Loss_L signal. This action ensures that the BBUs will discharge even when AC power is present.

4.20. Pulse load management mode

The PSU shall have the function of pulse load management, especially for GPU-based load possibly with repetitive peak pulse load. A PSU maximum input power limit shall be adjustable and defined based on specific system requirements.

PSU high-voltage bus capacitance shall be maximized and utilized for higher buffer energy of supporting the pulse power. In the case of depleting the bus capacitor energy, a dedicated Pulse_Mgmt_L signal could be toggled to request the BBU to share the pulse load.

Physical connections of the Pulse_Mgmt_L signals are illustrated in the figure in 3.18. Each PSU now has an output Pulse_Mgmt_L signal and all the Pulse_Mgmt_L signals in one PSU shelf are connected together and sent to a BBU shelf via two RJ45 connectors. The Pulse_Mgmt_L signal on the BBU shelf will be rerouted to BBUs and each BBU has an input Pulse_Mgmt_L signal. Upon detection of the pulse load and PSU has depleted the available bulk capacitor's energy, PSU would assert the Pulse_Mgmt_L signal to request load sharing from BBU. When detecting BBU's refusal of the load sharing, PSU shall release or increase the maximum input power limit to carry the load.

4.21. Pulse power requirements

The PSU should be capable of supporting the following repetitive peak pulse load while maintaining output within regulation and input power within the maximum limit.

≤140%, ≤50ms moving average
≤160%, 400us moving average
≤100%, ≥1s moving average.

BBU backup should not be triggered during the pulse loading operation.

Maximum output voltage under/overshoot should meet the requirement in Table 6.
The PSU input current waveform shall not be clamped during pulse loading operation.

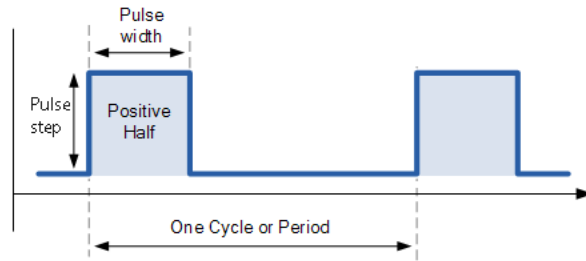


Figure 3: PSU Input Current Waveform

Pulse load profiles under test include but not limited to the ones shown in Table 9.

No.	Low load	High load	T_low	T_high	Average load	Frequency	load current slew rate
1	70%	130%	0.05ms	0.05ms	100%	10kHz	3A/us
2	20%	140%	0.1ms	0.1ms	80%	5kHz	3A/us
3	40%	160%	0.1ms	0.1ms	100%	5kHz	3A/us
4	40%	160%	0.4ms	0.4ms	100%	1.25kHz	3A/us
5	0%	80%	1ms	1ms	40%	500Hz	0.3A/us
6	20%	140%	50ms	50ms	80%	10Hz	3A/us
7	60%	140%	50ms	50ms	100%	10Hz	3A/us
8	30%	140%	28.6ms	50ms	100%	12.7Hz	3A/us
9	30%	160%	0.34ms	0.4ms	100%	1.35kHz	3A/us
10	30%	175%	2.14ms	2ms	100%	241Hz	3A/us

Table 9: Example Pulse Load Profiles

4.22. Output Over Voltage Protection (OVP)

The PSU shall shut down for DC output voltage exceeding 52.5V and the reacting time shall not exceed 200ms. The DC output voltage shall not exceed 54V (fast OVP).

4.23. Over Power / Current protection

If a PSU is overloaded higher than the values listed below in section 4.23.1, it shall shut down. 5s after an overload shuts down, if detecting 50V bus voltage is alive, PSU shall retry once to sync turn on. Otherwise, PSU shall latch off. During the one-time retry with 50V bus, if the fault is still present, the PSU shall latch off.

No AC_loss_L signal shall be asserted during overload condition to trigger BBU backup

The latch off can be cleared by AC cycling, reseating or a Modbus command. The OCP retry counter shall be cleared 30 minutes after normal operation.

4.23.1. Over Power/Current Protection Limits

PSU shall be self-protected for any overcurrent and short circuit conditions.

PSU shall set multiple protection levels and delay time for over current and short circuit protection properly. The OCP settings should be reviewed by all stakeholders. Below are the minimum limits required.

- Average power is more than 115% for 10s.
- Average power is more than 120% for 100ms.

4.24. Output Short-circuit Protection

The PSU shall employ short-circuit protection to protect the PFC and DC-DC circuits and attached load in the case of an output short-circuit or other output overload condition.

If the PSU output voltage is lower than 10V (short circuit condition), the PSU shuts off immediately. No component shall be damaged. PSU shall only retry once to sync start after 5s if detecting 50V bus voltage is still alive. Otherwise, PSU shall latch off.

No AC_loss_L signal shall be asserted during output short-circuit condition to trigger BBU backup.

The latch off can be cleared by AC cycling, reseating or a Modbus command. The SCP retry counter shall be cleared 30 minutes after normal operation.

4.25. Over Temperature Protection

The PSU shall employ over-temperature protection for both ambient over temperature and internal thermal temperature to protect the PSU. The PSU shall shut down under over temperature condition and recover after a certain period after the over temperature condition is removed.

The OTP circuit shall incorporate built in hysteresis such that the power supply does not oscillate on and off due to temperature recovery condition. The OTP event shall be reported as a fault.

4.26. Active Current Sharing Accuracy through analog bus

The PSU shall have a dedicated analog bus with slow bandwidth for active current sharing.

- With the maximum number of PSUs connected in the system, the current sharing accuracy shall be +/- 2% or better under load > 50% and +/- 5% or better under load >20%.
- Max setpoint increase (capture range) for the current regulation is 500mV.
- Current sharing bus (Ishare) voltage full scale shall linearly increase to 7V and scale up linearly up to 150% (10.5V).
- The PSU design shall be capable of supporting at least 10x power shelves in parallel operation. The PSU input impedance of the current sharing shall be 30kR & 100pF or better.
- Active current sharing and droop sharing response time should be minimized for cooperation with pulse power capacitor shelf. Active current sharing loop should meet the stability requirements.

- The active current sharing total delay time, measured from the end of load current variation to the end of Ishare out variation should be less than 50us.
- The active current sharing control bandwidth shall be less than 100Hz.

In the event that active current sharing malfunctions, the PSU shall share the load through voltage drooping with accuracy up to 5 times the values above. The failure of a module inside the PSU should not affect the load sharing or output voltages of the other supplies still operating. The supplies must be able to share load in parallel and be able to hot-swap.

4.27. PSU physical addressing

Three PSU signal pins are used for the slot physical addressing. These are digital signals that shall have internal pull up resistors inside the PSU. On the power shelf, these pins can be grounded (0) or left open (1) to determine PSU location as below:

PSU slot location 1: 000

PSU slot location 2: 001

PSU slot location 3: 010

.

.

PSU slot location 6: 101

For shelf design with more than 1-row PSUs, a 4th bit and signal pin might be defined.

4.28. LED

The PSU shall have a single blue and single amber LED mounted near the PSU handle for accessibility. Following are power supply LED States:

LED 1, Blue LED:

- 1) **Blinking Blue @ 4Hz frequency:** Sync Start State, PSU is ready to turn on its output and awaiting the sync Start signal
- 2) **Solid Blue:** 50V is ON and available
- 3) **No LED:** 50V output off

LED 2, Amber LED:

- 1) **Blinking Amber @ 4Hz frequency:** Bootloading
- 2) **Solid Amber:** Primary/Secondary/Fan/bootloading Failure and/or loss of DC output
- 3) **No LED:** fault NOT present/condition 1 and 2 are false.

- NOTE: toggling AC input power will reset the solid/blinking yellow fault light but will come up again if faults re-occur.
- Only one of the 3 conditions per LED will be applied at ALL times.

Refer to the mechanical drawing for the suggested location of the LEDs.

4.29. Grounding

The protective earth ground pins shall be connected to the safety ground (power supply chassis). This grounding should be well designed to ensure passing the max allowed Common Mode Noise levels.

PSU output voltage return shall not be connected to earth/safety ground.

PSU signal ground is directly connected to the output voltage return. A 10Ohm resistor is reserved between the Shelf/PMI/PMM signal ground and PSU signal ground.

Signal ground coming from TOR shall be isolated from the PMI/PMM signal ground through a 10 ohm resistor.

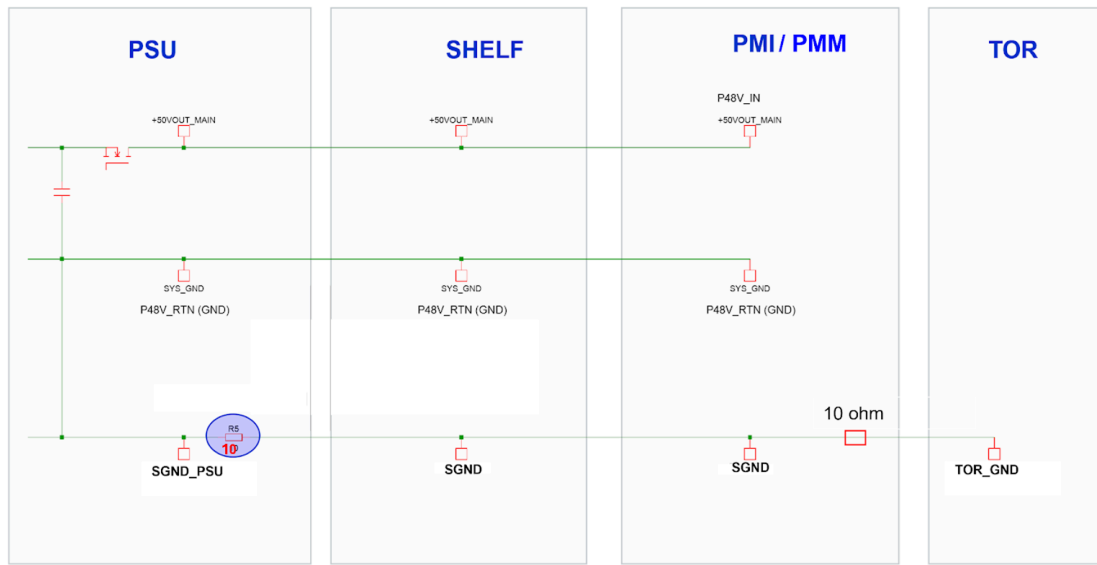


Figure 4: HPRv2 Grounding Scheme

4.30. Y-Cap

There should be no Y-caps placed outside the Oring FET between 50V and chassis. This measure is taken to prevent single faults, hot plug inrush currents, and static state leakage currents.

4.31. Random timer

Under any conditions of dissipative load, capacitive load, temperature, with or without backup voltage connected to the PSU:

- Max time for PSU to be “power-up ready” after AC voltage starts is 3s.
- After “power-up ready”:
 - When there is no dc voltage on the bus (first AC turn on) the power shelf shall be randomized with 0~2 second window to give each power shelf a random turn-on time (six PSU turn-on is synchronized).
 - When there is dc voltage on the bus higher than 44V for 0.1 sec (BBU is discharging), the power shelf shall be randomized with 0~5.5 second window to give each power shelf a random turn-on time (six PSU turn-on is synchronized).
- The power shelf shall turn on with only 1 PSU inserted into any slot.

Note: The random numbers above shall be dynamically generated immediately after each AC recycle, and not generated one time and then stored in the EEPROM for future usages.

Item	Description	Min	Max	Units
T_power-up_ready	Time for PSU to be power-up ready	1	3	seconds
T_random_noBBU	0-2 second initial turn on random delay without BBU discharging	0	2	seconds
TON_noBBU	Time 50VDC turns on after the shelf receives AC input without BBU discharging.	1	5	seconds
T_random_BBU	0-5.5 second turn on random delay after BBU discharging	0	5.5	seconds
TON_BBU	Time 50VDC turns on after the shelf receives AC input with BBU discharging.	1	8.5	seconds
TSYNC	After all PSUs in the shelf are ready to start till when 50VDC will start	2	5	msec
T-Max_ON_noBBU	Max PSU turn-on time without BBU in case of sync failure	6	8	seconds
T-Max_ON_BBU	Max PSU turn-on time with BBU in case of sync failure	9.5	11.5	seconds

Table 10: Random Timer**4.32. Transition and synchronization Requirements****4.32.1. Start-up (0V to 50V) Transition procedure**

Each PSU has a small circuit to output a SYNC_START signal and all PSUs' SYNC_START signals are connected together on the power shelf backplane.

- 1) Each PSU sets its SYNC_START signal to high when the PSU is ready to turn on the output.
- 2) Only PSU in SLOT #1 generates a random timer and sets SYNC_START to high when ready and the random timer is finished.
- 3) The PSUs will turn on the output when the SYNC_START signal is high.
- 4) If the SYNC_START signal is kept stuck low for 3s more than the max random timer limit (which is 2s without BBU and 5.5s with BBU), the PSU shall turn on immediately.
- 5) If PSU1 is not installed or its SYNC_START signal is stuck low, the other PSUs shall turn on within 10s. In this case a random timer doesn't exist.

Please refer to section 4.3 for the timing and constant current value requirements during the cold startup.

4.32.2. AC_Fault_CLR / Sync_Stop

Each PSU has a small circuit to output a dual-used AC_Fault_CLR / Sync_Stop signal and all PSUs AC_Fault_CLR / Sync_Stop signals are connected together on the power shelf backplane.

- 1) The "AC_Fault_CLR / Sync_Stop" signal is normally high with an internal 100k pull-up resistor and diode to 3.3V aux power.
- 2) Function to clear overload faults (OCP/SCP/OPP) after ac fault events
When detecting the AC becomes good after AC fault conditions, PSU asserts AC_Fault_CLR signal in the form of a 100ms negative pulse, that is High -> Low 100ms -> High and pulse window at 50ms to 200ms.
All other PSUs on the sync bus will clear the overload faults once detecting the AC_Fault_CLR signal and restart.
- 3) Function to synchronize the main output turn-off among PSUs

PSU sets Sync_Stop signal in the following conditions in the form of a 10ms negative pulse, that is High -> Low 10ms -> High and pulse window at 5ms to 20ms.

- a. Sync_Stop signal is asserted when PSU detects shelf output barclip OTP.
- b. Sync_Stop signal is asserted at the Power_Off_Unix_Time after the PSU receives an emergency off command (Power_Off_Unix_Time is set) from TOR.

All other PSUs on the sync bus will latch-off shutdown synchronously once detecting the Sync_Stop signal.

The Sync_Stop latch-off shutdown will be cleared with AC power cycle, module reseating, or system fault command received. AC_Fault_CLR signal shall not be able to clear the Sync_Stop latch-off shutdown.

4.33. Power system stability requirements

To be provided.

4.34. PSU connector pinout

Suggested PSU output connector: Amphenol PN 10169891-643002SLF (or equivalent).

More details can be found in section 8.7.

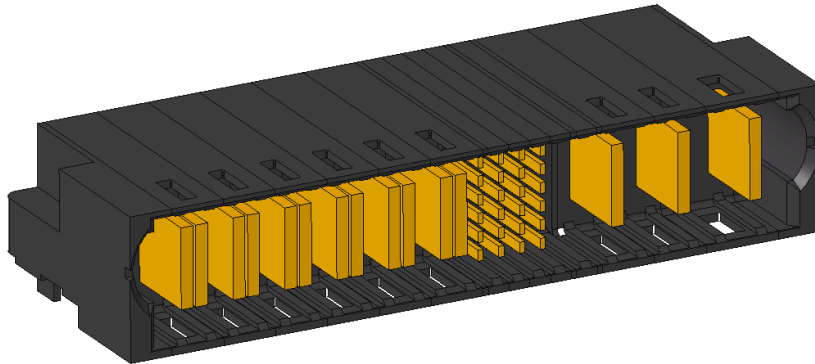


Figure 5: PSU Output Connector

P1 & P2 & P3	48V positive
P4 & P5 & P6	48V return
L1	Earth
L2	ac phase
L3	ac phase

Note: P4, P5, P6 and L1 mate first. D2 is a short pin.

PSU Pinout	Signal Name	Type	Bus	Function	Comments
F1	PSU_A0	Input		Address 0 – PSU ID A0	Internal pull up 10k to 3.3V
F2	PSU_A1	Input		Address 1 – PSU ID A1	Internal pull up 10k to 3.3V
F3	PSU_A2	Input		Address 2 – PSU ID A2	Internal pull up 10k to 3.3V
F4	Ground	Ground			
E1	Alert	Output/ Active Low	individual	Logic “Low” = Fault or Warning Logic “High” = OK	Open-collector/drain output, internal pull up 10k to 3.3V
E2	Reset_Latch_Fault	Input/ Active high	individual	high for 1-2s = clear faults and start PSU if not operating due to a fault	Should be enabled by SW. Internal pull down 10k resistor.
E3	Busbar_Clip+_Temp	Input		Busbar clip + contact temperature sensor	
E4	Busbar_Clip-_Temp	Input		Busbar clip – contact temperature sensor	
D1	AC_Loss_L	output/ Active low		Low: AC loss	Open-collector/drain output, internal pull up 100k and a diode to 3.3V
D2	PSKILL (Short Pin)	Input/ Active High		Logic “Low” = Output Turn on Logic “High” = Output Turn off	Quick shut down Output, mitigate hot unplug arcing. Internal pull up 10k to 3.3V.
D3	RS485A	BI	Share bus		

D4	RS485B	BI	Share bus		
C1	Pulse_Mgmt_L	Output	Active Low	Logic “High”= no request for BBU support of pulse load Logic “Low”= request BBU support for pulse load	Open-collector/drain output, internal pull up 100k and a diode to 3.3V
C2	Shelf_addr_In t_A0	Input	Share bus	ignored	Internal pull up 100k to 3.3V
C3	Shelf_addr_In t_A1	Input	Share bus	ignored	Internal pull up 100k to 3.3V
C4	Shelf_addr_In t_A2	Input	Share bus	ignored	Internal pull up 100k to 3.3V
B1	ISHARE	Analog	share bus	Main Output current share bus	
B2	I2C_SDA	BI	share bus	I2C Data	
B3	I2C_SCL	BI	share bus	I2C Clock	
B4	Ground	Ground	share bus	I2C ground	
A1	Present_L	Output	Individual	Low: PSU/BBU present	1. 10Ω pull-down inside PSU/BBU module. 2. 4.7k pull-up inside PMM. 3. Connecting to PMM edge connector pin A27/B27/A28/B28/A29/B29 respectively.
A2	AC_Fault_CLR / Sync_Stop	BI	share bus	clear latch off flag after ac fault event	Open-collector/drain output; Internal pull up 100k and a diode to 3.3V
A3	Sync_Start	BI	share bus	Synchronizing turn-on main output	Open-collector/drain output, Internal pull up 100k and a diode to 3.3V
A4	Ground	Ground			

Table 11: PSU Connector Pinout

Note:

1. The pull up resistor of all the shared signals should be individually in series with a Schottky diode, so the problem in one PSU doesn't impact other PSUs.
2. Logic signals level (3.3V high and 0V low) measured on both PSU-level and shelf-level should meet the requirements of $\geq 3V$ for high and $\leq 0.3V$ for low in different operation conditions.

4.35. Interoperability

PSU module to shelf interoperability shall be ensured, that is PSU modules from vendor A shall be able to operate normally in the power shelf from vendor B, and vice versa.

All PSU modules on the same rack must be from the same supplier.

5. Communication

The PSUs shall be able to communicate on ModBus (up to 115kbps).

The software interface shall be operational when the AC is present or when the DC output bus is powered up by other power sources. Refer to the OR v3 HPR v2 PSU Modbus register maps for fault/read/write registers that PSU should communicate.

5.1. Firmware Upgrades

The interface shall allow the user to re-flash firmware on the device. Firmware upgrade shall result in no power interruption on the shelf level (the unit being upgraded can go offline.) Upgrades can be done one PSU at a time. PSU output voltage interruption due to FW upgrade shall be less than 5s.

The PSU FW shall maintain regulation on the output during Send, Install and verification of the new FW, and only require a soft reset (that may reset the output for a short period in a few seconds)

Suggested implementation

Traditional designs have the following implementation on its micro flash at a high level:

Upon micro boot, when FW update is not enabled, i.e., the unit is not being updated, the unit boots into the application mode and carries out the operation.

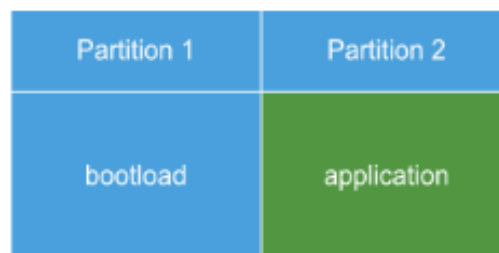


Figure 6: Traditional Firmware Partition Layout

As an update is initiated, the bootloader partition is leveraged for the duration of the update, the new FW is then installed in the application partition, a soft reset occurs, and unit boots into the newly installed application

In the improved method, the microcontroller flash is split into three partitions: a bootloader partition and two application partitions.

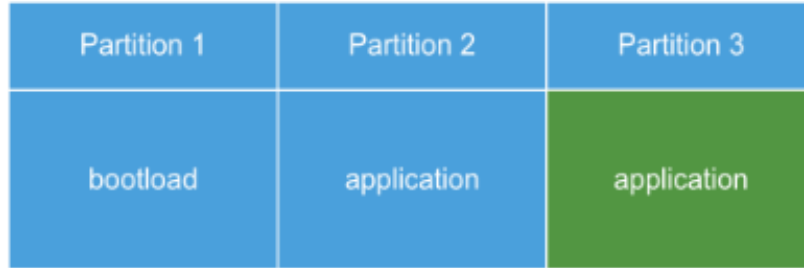


Figure 7: Improved Firmware Partition Layout

During normal operation, the unit runs from one of the two application partitions. The firmware update process also remains the same. However, unlike the traditional method, the new firmware is written to the application partition that is not currently running. Once the installation is complete, the unit resets and boots into the newly installed image. This approach has the benefit of always maintaining a last-known-good copy of the firmware on the PSU.

Note that this implementation depends on the architecture of the microcontroller used, in some cases, 3 separate partitions are not required, and a similar process can take place with two partitions in the flash and the presence of a ROM.

5.2. Metering Accuracy

Accurate reporting of input power (power factor, input current, input current harmonics and voltage) and output power (output current and voltage) readings shall be reported via communication system at all rated voltages.

The accuracy shall be maintained across the operating temperature range and between 200Vac and 305Vac.

Parameter	Load	Accuracy
AC input power	<10%	±25W
	10-20%	± 5%
	20-100%	±3%
AC input current	<10%	±0.5A
	10-20%	±2%
	20-100%	±1%
AC input current THD (Error difference not %)	<10%	±10%

	10-20%	±2%
	20-100%	±1%
Power factor (Error difference not %)	<10%	±0.1
	10-20%	±0.02
	20-100%	±0.01
AC input voltage	0-100%	±1%
Output voltage	0-100%	±0.5%
Output current	10-20%	±10%
	20-50%	±5%
	50-100%	±1%
Output power	<10%	±25W
	10-20%	±3%
	20-100%	±2%

Table 12: PSU Metering Accuracy

5.3. Blackbox Function

The black box function shall store key important data to be used when a fault occurs.

- Must store data in memory and be able to withstand several read/write cycles.
- PSU must be able to store failure data before the PSU turns off/fails even in catastrophic failure events both on primary and secondary side. Hold up time of the blackbox microcontroller must be able to store all the information and then shutdown.
- Last 4 events stored in memory.
- AC input current, AC input voltage, Input Power, Power factor, iTHD, DC output voltage, DC output current.
- Temperature readings, fan speed, input voltage, output voltage, bulk voltage, various error codes from all the different converters (OTP, OVP, OCP,UVP), and warnings.
- Total run time of PSU.
- Run time since last turn on.
- Real time stamping.
- Number of AC power cycles.
- Number of AC outages (can be determined by going into backup without counting the battery test conditions).

Power supply event data is saved to the Black Box for the following events:

Any events that caused the Main Output to shut down:

- Main Output over voltage fault.
- Main Output under voltage fault.
- Main Output over current fault.

- Main Output short circuit fault.
- Fan failure.
- Over temperature fault.

Any events that caused the AC input to be bad:

- AC Input under voltage fault.
- AC Input over voltage fault.
- AC Input out of range frequency fault.

5.3.1. Fault Log History

The FAULT_LOG_HISTORY is used to read the event data of the unit. The fault log history data stores up to 4 event data sets. If the event data sets exceed 4, the oldest data set shall be removed to give way for the latest data set. Unused history data sets shall have a value of zero for all registers.

6. Environment

6.1. Temperature

- Operational or cold aisle (inlet) temperature: -5°C to +45°C
- Non-operational: -40°C to +85°C

6.2. Humidity

- Operational : 10-90% RH non-condensing
- Non-operational : 5-93% RH non-condensing

6.3. Gaseous Contamination

- Sample Size 1
- EIA 364-65 IIA
- 30°C, 90%RH, 10 ppb H₂S, 200 ppb NO₂, 10 ppb Cl₂ and 100 ppb SO₂.
- Duration: 12 days for mated condition and 12 days for un-mated condition.

6.4. Altitude

- Operational : 0-3050m
- Non-operational : 0-12000m

6.5. Acoustic Noise

- Target sound pressure should not exceed 85dBA when fan modules are running at full speed and operating within the defined environmental envelope

6.6. Vibration and Shock (non-packaged)

The PSU shall meet vibration and shock tests per EN 60068-2-6 and 60068-2-27, respectively, for both non-operating and operating conditions, with the specifications listed below. The test should be performed inside the power shelf with all 6 PSUs and PMI installed. During operating vibration and shock tests, the PSU shall exhibit full compliance to the specification without any electrical discontinuities. During the non-operating tests, no damage of any kind (including physical damage) should occur and there should be no functional impact of the PSU per the specifications.

Shock and Vibration tests are to be performed on soft tooled and hard tooled parts.

Vibration Non-Operating:

Endurance

Excitation Mode	Sinusoidal
Test Frequency	5Hz to 500Hz
Amplitude	(5.0-9.0Hz) 6mm peak to peak (9.0-500.0Hz) 1g
Frequency Change Rate	1 octave / min
Test Directions	3 directions in space (x,y,z)
Duration	10 sweep cycles for each direction (2hours 13 minutes)
Test Temperature	Room temperature
Electrical Work	None

Functional | per MIL-STD-810, Method 514 (random vibration):

Excitation Mode	Random
Test Frequency	5Hz to 500Hz
Overall Level	1grms
Test Directions	3 directions in space (x,y,z)
Duration	1 hour / axis (3 hours)
Test Temperature	Room temperature
Electrical Work	None

PSD Profile:

Frequency (Hz)	PSD (g ² /Hz)
5	0.000238
20	0.00238
80-350	0.00238
500	0.000238

Shock Non-Operating:

Shock Pulse:	half sinus
Shock duration:	11ms
Shock Amplitude:	12g
Test Directions:	6 directions

Number of Shocks:	60 (10 per each direction)
Test Temperature:	Room temperature
Electrical Work:	None

Vibration Operating:

Endurance

Excitation Mode	Sinusoidal
Test Frequency	5Hz to 500Hz
Amplitude	(5.0-9.0Hz) 6mm peak to peak (9.0-500.0Hz) 1g
Frequency Change Rate	1 octave / min
Test Directions	3 directions in space (x,y,z)
Duration	10 sweep cycles for each direction (2hours 13 minutes)
Test Temperature	Room temperature
Electrical Work	Power supply in operation

Functional | per MIL-STD-810, Method 514 (random vibration):

Excitation Mode	Random
Test Frequency	5Hz to 500Hz
Overall Level	0.5grms
Test Directions	3 directions in space (x,y,z)
Duration	15 minutes / axis (45 minutes)
Test Temperature	Room temperature
Electrical Work	Power supply in operation

PSD Profile:

Frequency (Hz)	PSD (g ² /Hz)
5	0.0000594
20	0.000594
80-350	0.000594
500	0.0000594

Shock Operating:

Shock Pulse:	half sinus
Shock duration:	11ms
Shock Amplitude:	6g
Test Directions:	6 directions
Number of Shocks:	30 (5 per each direction)
Test Temperature:	Room temperature
Electrical Work:	Power supply in operation

6.7. Package Vibration, Drop and Compression

PSU units in their package shall meet the following requirements:

Package Vibration	1.146 G _{rms} , 2-200-2 Hz, all three axes, Random Vibe	ISTA 3E 06-06
Package Drop	8-inch drop	ISTA 3E 06-06
Package Compression	Maximum compression loading on a bulk pack	ASTM D 642-94

7. Thermal

7.1. Operational

7.1.1. Airflow Direction

Front-to-back.

7.1.2. DeltaT

This is the air-side temperature difference across the PSU assembly and represents airflow usage (CFM) in relation to heat loss in the device (W). Efficient use of airflow is critical to meeting facility-level requirements, and the following temperature difference targets are outlined.

Air inlet and outlet temperature difference should be higher than **12°C** across 30~85% load range and up to 35°C inlet/ambient and 3050m (10,000ft) above sea-level for steady state loads.

Air inlet and outlet temperature difference should be within the range of **12 to 25°C** across 85 to 100% load range and up to 35°C inlet/ambient and 3050m (10,000ft) above sea-level for steady state loads.

7.1.3. Thermal Margin

While efficient use of airflow is important for thermal design, reserving adequate margins on components is equally critical. These margins should be defined with respect to de-rated values, as appropriate. Requirements are as follows.

Component thermal margin of $\geq 7\%$ or $\geq 5^{\circ}\text{C}$ under worst-case condition (100% load, input voltage lowest, output voltage lowest, with GPU pulse load) and **up to** 30°C inlet/ambient and 3050m (10,000ft) above sea-level. Target whichever margin value is larger.

Component thermal margin of $\geq 4\%$ or $\geq 3^{\circ}\text{C}$ under worst-case condition (100% load, input voltage lowest, output voltage lowest, with GPU pulse load), at **greater than** 30°C inlet/ambient and up to 3050m (10,000ft) above sea-level. Target whichever margin value is larger.

Margin to de-rated temperatures should account for associated differences in reading and measurement location. Impact on reliability should also be considered when determining required margin.

7.2. Thermal design requirements

7.2.1. Sensor Accuracy:

For discrete and critical sensors (such as ambient temperature), an accuracy of $\pm 2^{\circ}\text{C}$ is required ($\leq \pm 1^{\circ}\text{C}$ is preferred). Exhaust and inlet temperature sensors are required and calibrated in the unit. If a component does not have an integrated temperature sensor, and uses a proxy, it needs to target an accuracy $\leq \pm 5^{\circ}\text{C}$ ($\leq \pm 2^{\circ}\text{C}$ is preferred). If this component is temperature sensitive, thermal margin requirements defined above should account for sensor inaccuracy.

7.2.2. System Fan

Should be sized to support operation across environmental and loading envelopes, with an adequate operating range (speed) to achieve requirements outlined in this document. The fan should also have adequate overhead to accommodate back-pressure resulting from shelf design, rack-level accessories and data center operation. In general, head room to overcome a back-pressure of ≥ 0.3 inches of water is highly recommended, and 0.2 inches of water is a must-have. When the PSU is plugged in, the fan should stay operational irrespective of load and maintain front-to-back airflow direction (overcome potential back-pressure). Mounting of the fan must meet any vibration and acoustic criteria and should not violate any physical constraints outlined. The fan should be included within the power supply enclosure.

7.2.3. Surface Temperature

To make the PSU assembly safe for handling in-operation, accessible surfaces should not exceed a temperature of 75°C with full load and nominal conditions (nominal VAC, 35°C and 6000 ft).

In case the rear side surface temperature is higher than 75°C with worst pulse loads and nominal conditions, a high temperature warning label should be added on the top cover of the shelf at the rear side.

7.2.4. Rear Side Clearance

The thermal design is able to support the cooling with **max 1-OU air blocker** in the rack level structural design to be installed on the rear side of the power shelf with **minimum 70 mm** distance from the rear panel.

7.3. Fan Failure

If a fan fails, the PSU must indicate the failure with a signal that will be reported via software as well as an LED indicator on the front panel. The PSU shall not need to shut down because of a failed fan and only shut down if there is a fault, i.e. over-temperature fault.

7.4. PSU Thermal Monitoring

Each PSU shall provide the following parameters via the defined communication protocol (in addition to monitoring of PSU components and operating parameters). The following thermal parameters must be available for each PSU and labeled accordingly:

- Inlet temperature
- Exhaust temperature
- Fan RPM (average, if >1 fan used in the PSUs); percent is acceptable as long as full speed rpm is provided at some point
- Fan fail signals

7.5. Shelf Output Busbar Clip Temperature Monitoring and Protection - if Busbar Clip is Selected

- There are two thermistors attached to the busbar clip. One thermistor is in contact with the positive (+) metal contact, and the other thermistor is in contact with the negative (-) metal contact. The thermistor should meet the sensor accuracy spec for all operating conditions.
- The PSU shall report the temperatures through Modbus registers.
- The PSU shall set an alarm bit within a Modbus register if the temperature exceeds the warning threshold. This alarm bit serves as an indication that the temperature has reached a level that requires attention. Additionally, if the temperature exceeds the safety threshold, the PSU will initiate a shutdown process to ensure the safety of the system.

8. Mechanical

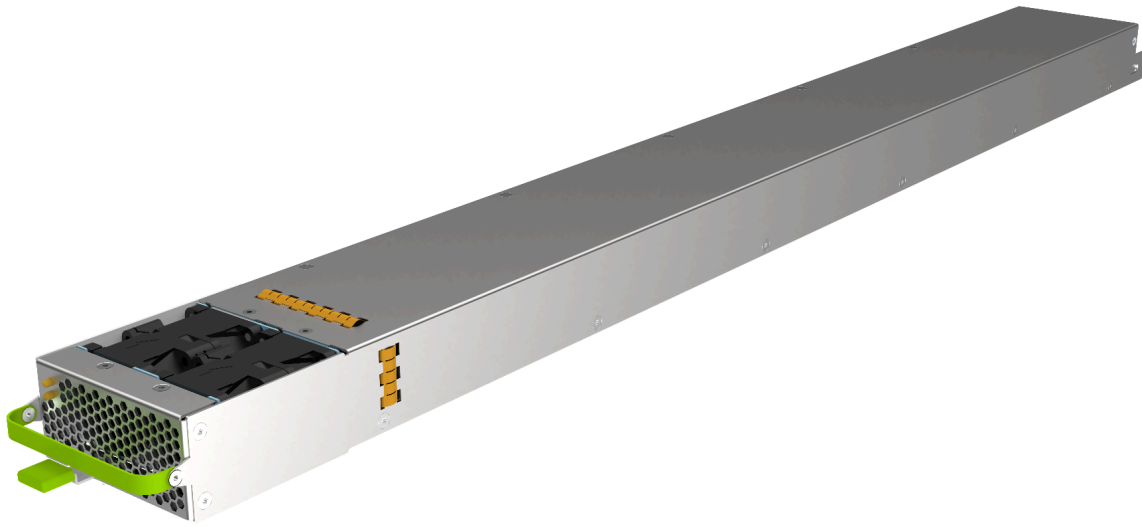


Figure 8: PSU Isometric View

8.1. PSU Physical Dimensions

The PSU size is 83.4mm x 40mm x 790mm [Width x Height x Depth]

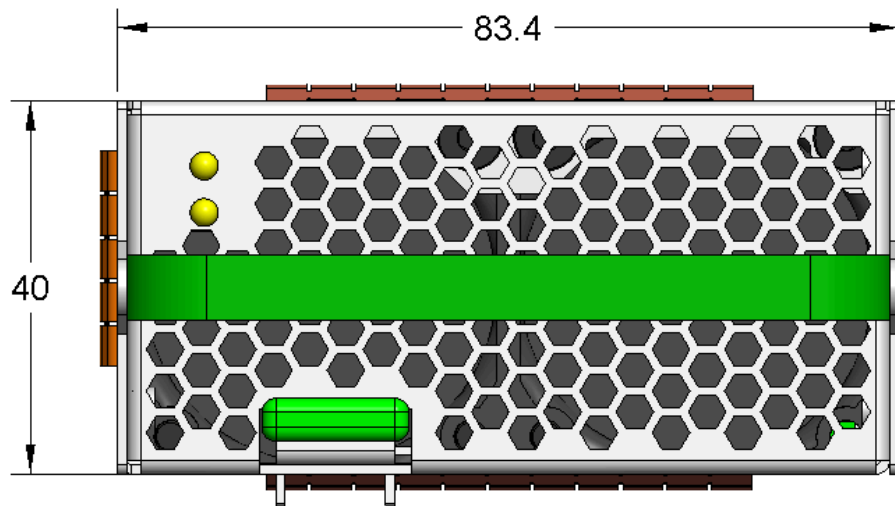


Figure 9: PSU Front View with Critical Dimensions

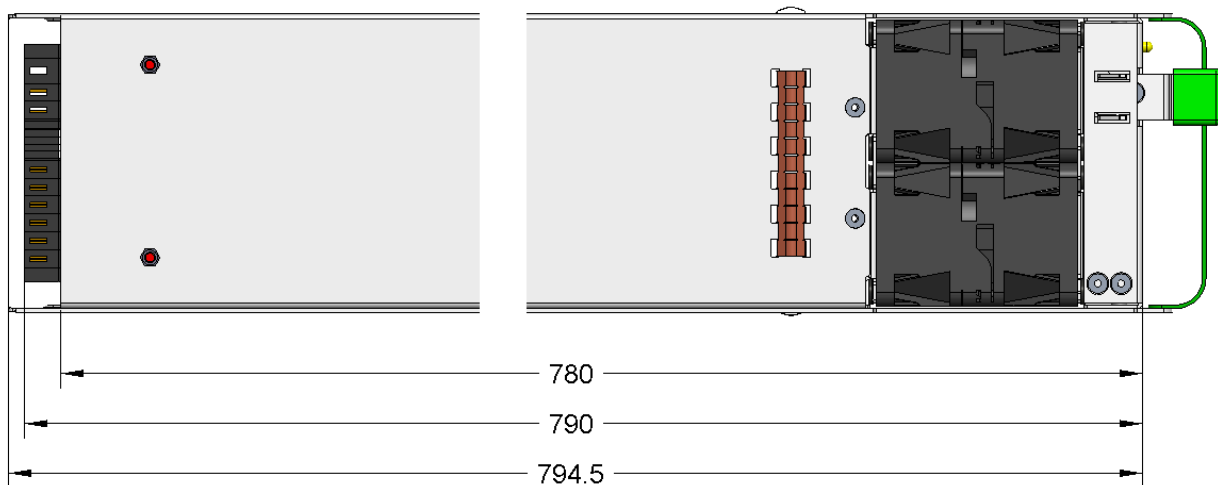


Figure 10: PSU Top View with Critical Dimensions

8.2. Construction

The PSU base and cover shall be assembled using flathead screws. No rivets are allowed as the PSU must have the ability to be opened using a screwdriver. The individual base and cover can be welded, riveted or screwed together, consistent with meeting shock and vibration requirements. There shall be no sharp corners or edges.

8.3. Materials and Fasteners

The sheet metal material shall be steel, pre-plated hot-dip zinc coated, with 0.8mm - 1mm of thickness unless otherwise specified. Any plastic material used should meet UL 94-V0 specifications. The system shall contain a minimum of 40% post-consumer-recycled (PCR) plastic content and 15% recycled steel content. However, functionality and reliability are paramount. 65% recycled PC/ABS and 20% recycled SGCC shall be qualified and used in applicable parts to meet the overall recycled content requirement. The following PCR plastics have been qualified for use:

- GLite MBS-200BKR01
- GLite MBS-200GMR01
- Kingfa JH960-6950 C2B-S0759
- Kingfa JH960-6950 C7G-S0023
- Wistron WAM NCT50T#

8.3.1. PCR Targets & Implementation

- PCR content must be met in aggregate for polymers and steel.
- Functionality and reliability take priority; waivers may be considered for components where recycled content impacts performance.
- PCR requirements are mandatory at MP; strongly recommended to pilot in DVT for risk mitigation.

8.3.2. Supplier Qualification & Material Testing

- All new material suppliers or materials not previously validated must complete full qualification testing to confirm performance and compliance with recycled content requirements.
- Suppliers already validated for the specified recycled content do not require requalification.
- Required testing (as applicable):
 - Mechanical: Tensile, hardness, standoff mounting strength.
 - Corrosion/finish: Salt spray, silkscreen print adhesion.
 - Sheet metal feature limits: Max emboss height, min L-flange bend height, min closed hem length, min inner hole radius of closed hem, and other relevant features.
 - Key mechanical properties must be tested to relevant JIS standards (or equivalent), or any other applicable standards.

8.3.3. Documentation & Certification

- Required
 - Recycled Content Certificate from supplier.



Figure 11: Example of RC Certificate

- Material Certificate showing compliance with relevant standards (e.g., JIS G 3302:2019).
- Metal coating certificate (e.g., Z18–Z22).
- RoHS compliance certificate.
- BOM with detailed breakdown: material, PCR %, aggregate PCR % by weight for polymers and steel.
- Optional but Preferred
 - ISO 17065/14021 accreditation.
 - Recognition from LEED, EPA, or BREEAM.

8.4. Latch & Handle Requirements



Figure 12: PSU Handle & Latch

A latch and handle are required for PSU removal. The latch shall be attached in the location shown on the mechanical drawing and is to interface with the cutouts on the power shelf chassis. The latch design may vary, but the finger interface of the latch must be Pantone 375C (Green).

A foldable handle is highly recommended. The handle shall be designed such that it does not protrude past the top and bottom surfaces when folded 90°. Handle touch points shall be Pantone 375C (green). Please note that the PSU will be heavy, and the handle should be sturdy enough to carry the entire weight of the PSU.

8.5. EMI Gaskets

EMI gaskets are to be placed on the left & right and/or top & bottom sides of the PSU. The EMI gaskets serve the following two purposes:

- PSU module-to-shelf grounding
- EMI containment.

Gaskets are to be placed towards the front of the PSU module so they make full contact with the walls of the PSU shelf.

Suggested placement distance from front panel along length of PSU module:

Right/Left Side: ~100mm

Top/Bottom Sides: ~100mm.

8.6. Chassis Interface

The handle of the PSU may not protrude from the front surface of the PSU by more than 23mm. The latch shall be located on the bottom side of the PSU and interface with a rectangular hole in the chassis. Please refer to the 3D drawing for latch location on the PSU. The latch shall only engage after proper electrical contact is made with the rear blind-mate connectors.

8.7. Rear Blind Mate Connector

The rear blind mate connector of the PSU shall be Amphenol 10169891-643002SLF or equivalent. This is a R/A plug connector with 6 high power pins, 24 signal pins, and 3 low power pins. Please refer to the drawing for more details.

The connector position within the PSU is fixed in the X, Y, and Z directions according to the 3D drawing. This cannot be altered due to mix-and-match requirements for the power supplies into the shelf.

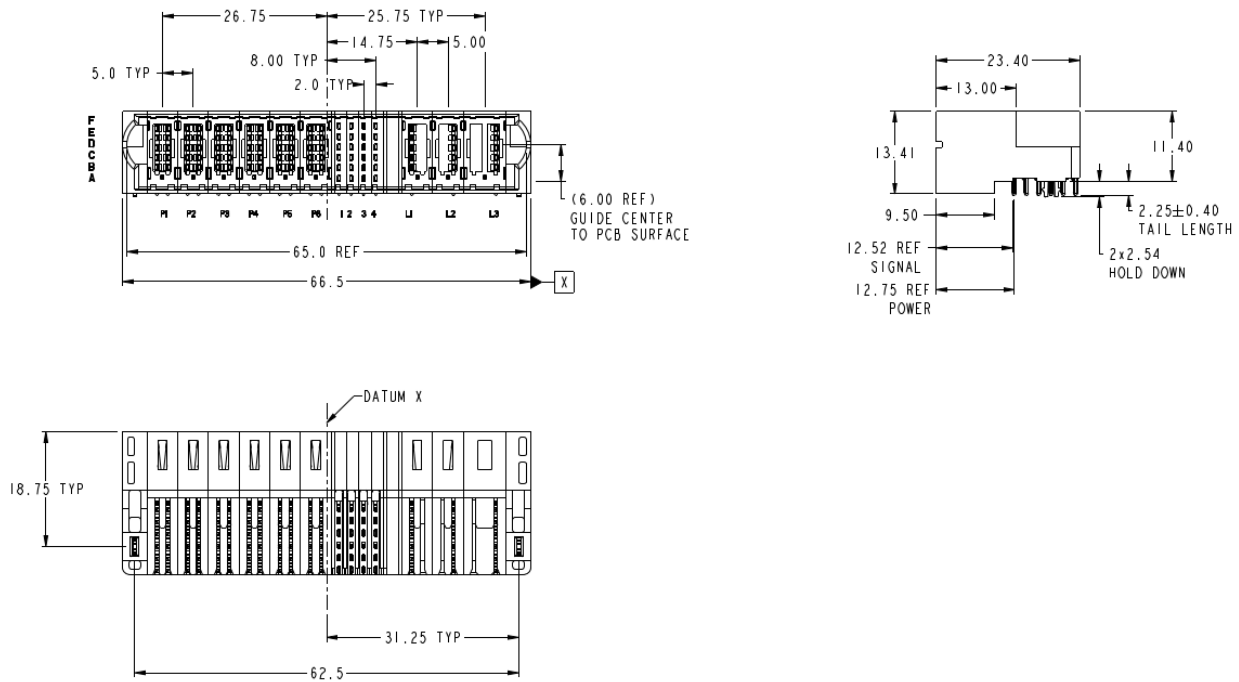


Figure 13: Rear Blind Mate Connector Drawing

8.8. Rear Blind Mate Connector Protection

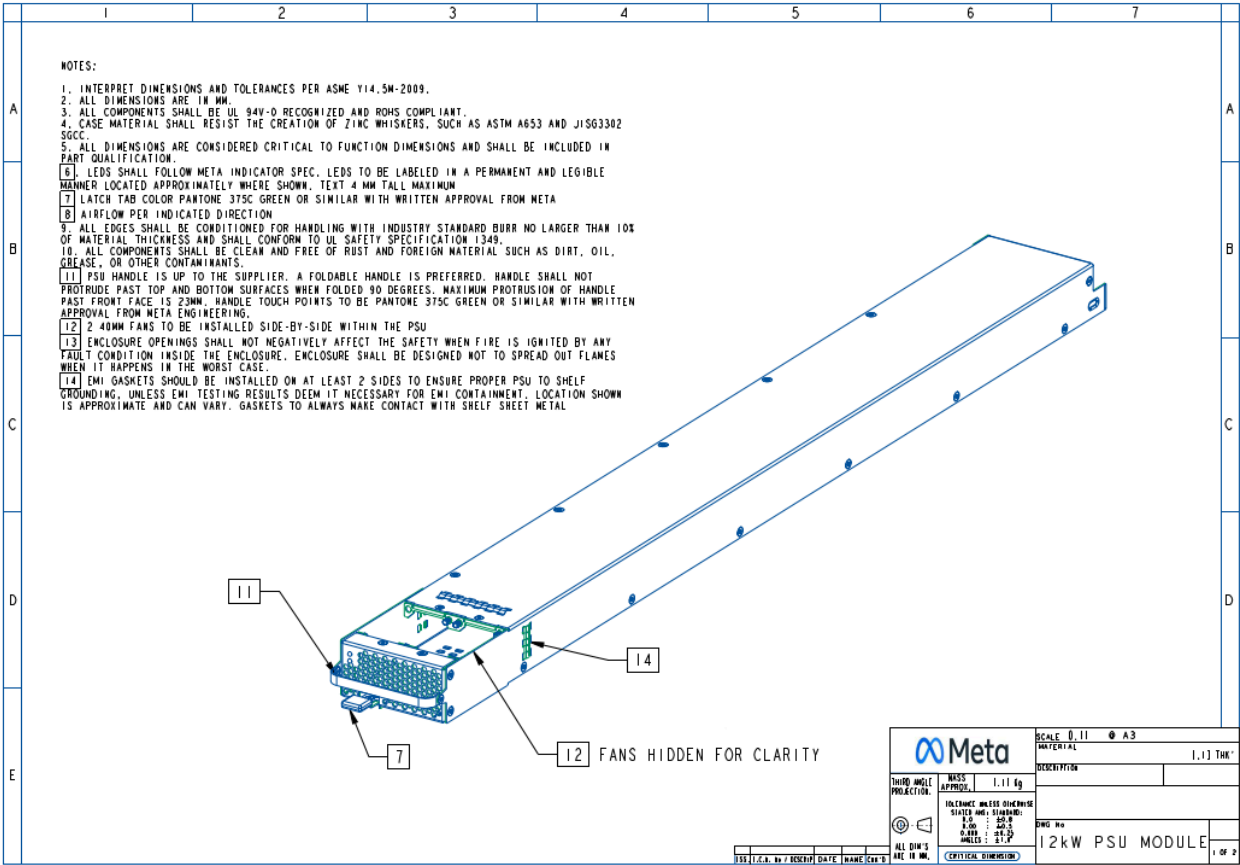
A method of protecting the rear blind mate connector on the PSU is required to prevent damage from accidental mishandling during manufacturing or end user handling of the module. Please review mechanical 3D CAD for reference design of protection tabs. These tabs should not interfere with any features on the PSU Shelf.

These tabs also double as a keying mechanism to prevent the PSU from being installed upside down.

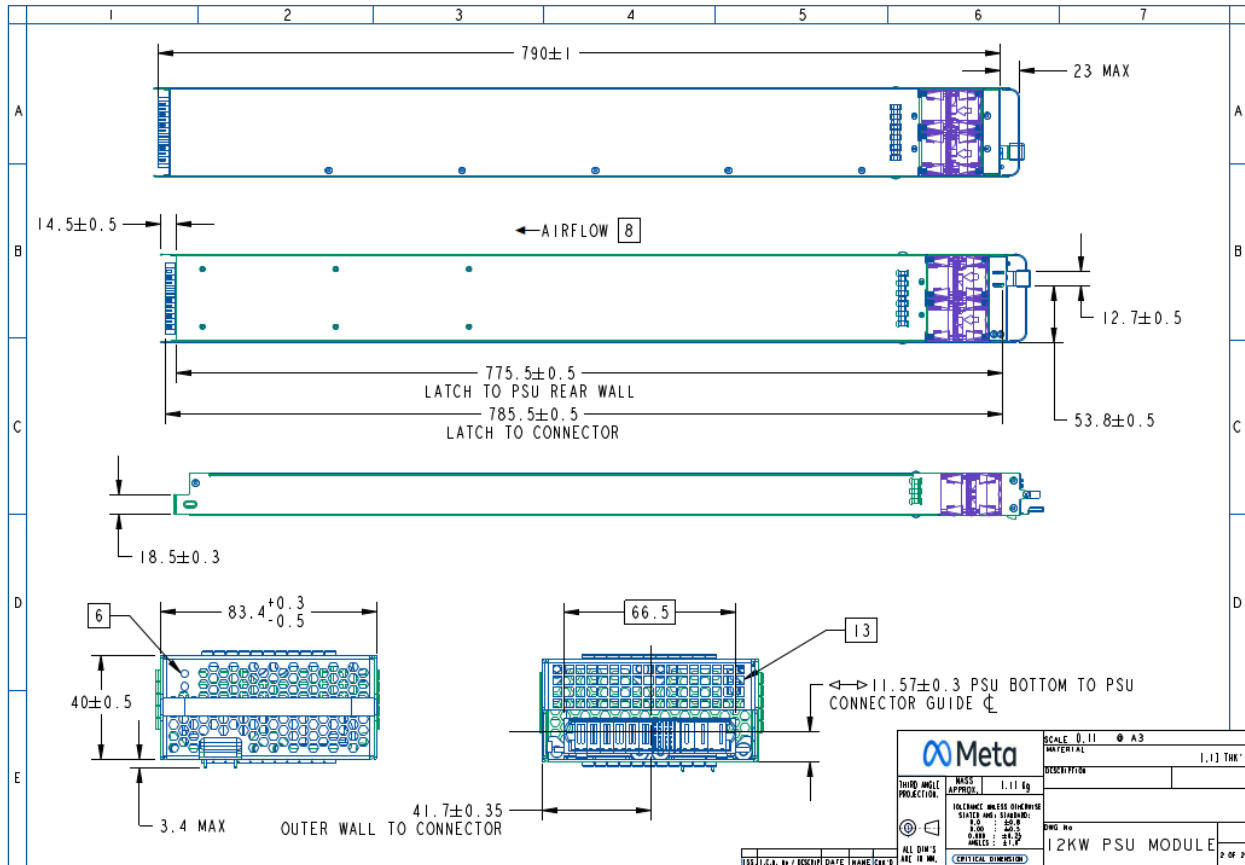


Figure 14: Rear Blind Mate Connector View

8.9. Mechanical Drawings



Open Compute Project • Open Rack V3 HPR V2 12kW PSU Specification



9. Reliability and Quality

9.1. Derating Design

- Component stress analyses must include Thermal and Electrical stress analyses, and the analyses must consider the worst case operating Thermal and Electrical conditions (defined in [Section 6 Environmental Requirements](#) and [Section 4 Electrical Requirements](#) of this document respectively) applied to all applicable components in the PSU module; Capacitor life must be included in the overall analysis as well when applicable.
- IPC-9592B may serve as a reference document for defining the derating specifications of various components, and all components must meet or exceed the specifications defined in IPC-9592B (Class II) as a minimum requirement.
- All Capacitors must meet/exceed > 6 years operation life requirement.
- This activity must be completed at each phase of NPI development (both Primary and Mixed Source, if applicable).

9.2. MTBF Requirements

- Vendor shall provide Calculated MTBF using Telcordia SR-332 latest version (Method 1 Case 1/Parts Count in EVT and Method 1 Case 3/Parts stress in DVT). Target MTBF is 500K hours and calculation should be at 35°C 6000ft elevation ambient temperature, 277Vac of input voltage, and 100% load. The fan is not included (MTBF of the fan will be provided separately).
- The PSU shall meet a demonstrated MTBF of minimum 500K hours at 90% confidence level prior to first customer shipment (Pilots samples, Mass Production units). DMTBF is to be performed with PSUs installed in the shelf using PSUs and power shelves from mix-source build.
- MTBF goal: the Vendor shall provide to customer the best MTBF numbers that the PSU will be able to meet, no matter the minimum requested. Customer will use this information for an overall reliability study of the whole system.
- The fan L10 life shall be at least 6 years at 35°C 6000ft elevation inlet air temperature and full speed (to be verified with the fan vendor)

9.3. Design Failure Mode and Effect Analysis (DFMEA)

- The risk assessment must be performed with a focus on the long term exposure to environmental, electrical, mechanical stresses for the lifetime of the product, as well as based on the design and worst case product operational requirements that are outlined in this document.
- The assessment must start as soon as a design of any of the critical elements is available for assessment.
- The risk assessment must include failure modes, impact to the functional/appearance of the system, failure occurrence that must be based on historical data or simulation results.
- Risk levels that are classified as High and Medium risks must have mitigation plans in place before Design Freeze of next stage of development, or at least by Phase Exit.
- Whenever there is a change in the design and/or operational requirements that affect the reliability of the PSU module, reassessment must be completed.
- Other areas of discipline (e.g. Quality, EMI/Compliance, Safety...) should be performed and reported separately.

9.4. High Accelerated Life Test (HALT)

- A comprehensive HALT shall be performed on the PSU module + PMM. The HALT equipment, testing procedure, sample size, testing report and documentation, and root cause analysis and corrective action requirements shall follow the requirements as specified in IPC-9592B, Section 5.2.3 and Appendix D.
- HALT test is to be performed on PSU samples from mix-source build.

9.5. NPI Reliability Tests

- The PSU module must pass a set of reliability tests that comprises environmental stress (defined in [Section 6 Environmental Requirements](#) of this document), electrical stress (defined in [Section 4 Electrical Requirements](#) of this document), and mechanical stress (operational/non-operational/transportation) conditions. IPC9592BL may serve as a baseline reference for the reliability test plan development, but the exact test item and the respective test conditions should be defined based on the product specifications and DFMEA results.
- Reliability tests must be performed at each phase of the NPI development starting from EVT, and continue in DVT and PVT.
- Test durations for some of the test items must meet or exceed > 6 years of PSU operation life. The acceleration factors that are based on thermal, electrical, etc may be employed to reduce the

test duration for the purpose of demonstrating the robustness of the design under the most extreme operational requirements as defined in this document.

- Sample types include PSU module + PMM + Shelf

9.6. Disallowed Components

The customer reserves the right to select some preferred/mandatory parts, during the development phase (selection of an approved vendor list (AVL) subset). The Vendor will work with the customer to solve system integration problems, should any issues arise.

- Trimmers and/or potentiometers.
- Tantalum capacitors.
- Dip switches.
- High side driver ICs
- Paralleled power MOS are allowed provided that the design prevents parasitic oscillations.
- SMT ceramic capacitors are allowed with the case size < 1206. The size 1206 can still be used when SMT capacitors are placed far from the PCBs edge, and with a correct orientation that minimizes risks of cracks [any size > 1206 must be highlighted to customer during DFM review].
- Allowed ceramics materials for SMT capacitors are: X7R or better material. The COG or NPO types should be used in critical portions of the design, such as feedback loop, PWM clock settings, etc. [any changes shall be highlighted and brought up during DFM review].
- Relays: the use of any electro-mechanical relays, and type shall be discussed up front before any approval is given to include them in the design.
- Resistors by Royal Ohm are not to be used.
- Customer will approve all control ICs.

9.6.1. Capacitors

- The PFC high-voltage bus electrolytic capacitors shall be rated at 105°C, no less than 475V and shall be selected from Japanese and US manufacturers only.
- All capacitors shall have a lifetime of at least 6 years at 35°C at 6000ft elevation inlet air temperature.

9.6.2. Sheet Metal Material and Zinc Whisker Implications

- Sheet Metal Chassis Material is hot-dip Zinc coated, JIS G3302
- SGCC (Z18 to Z22), with 0.8mm - 1mm of thickness. The 'Z' parameter defines the metal coating thickness: Z20 is for 40µm of thickness, and Z22 is for 43µm.
- The Japanese standard is 'JIS G3302', while the US standard is 'ASTM A653'.
- Mechanical design shall prevent sharp edges and possible metal oxidation in the critical points of the sheet metal (e.g. in the cut & bends portions, etc.).
- Both chassis design and metal base material shall not promote the growth and propagation of zinc and tin whiskers.
- Metal base materials with electro-zinc plating, or poor conductivity plating, are not allowed.
- Alloy materials are a possible option, while stainless steel is another possibility provided that it makes cost sense (both options are subject to customer approval).
- Aluminum material is not allowed for the enclosures.
- The chassis enclosure, as well as all electronics, shall meet certain contamination requirements (see EIA 364-65 IIA).

9.6.3. Mix-source Builds

Suppliers are required to conduct a mix-source build ideally at DVT using prime and 2nd source/alternate parts in the BOM. These samples are to be used in all Reliability and Environmental tests as agreed and approved by all stakeholders.

10. Manufacturing Quality Control and Testing

It is required to meet the quality process requirements as specified in IPC-9592B, Section 7 ("Quality Process"), which include PFMEA, statistical process control (SPC), corrective action process, yield control, materials traceability, product change notice (PCN), qualification of change, etc.

A PCN shall be submitted for any change that affects form, fit, function, compliance, revision, version, process, material (including deletion), testing (i.e. BI reduction) and other modifications. If uncertain, please contact the customer. The PCN shall be issued with an assigned unique number for traceability and formally submitted via email. The supplier shall give a minimum of one month's notice before target implementation of the proposed changes unless urgent (i.e. resolve design, quality, process or supply issue, customer request). PCN approval is required before implementation of any changes.

The PCN document shall include the following:

- a. FBPN and revision (from and to, retain if not changed)
- b. Vendor PN and revision/version (from and to, retain if not changed)
- c. Vendor PN HW and FW (if applicable) (from and to, retain if not changed)
- d. Date of Issuance
- e. Reason of change
- f. Description of changes
- g. Form, fit, function changes (if any)
- h. Before and after drawing with mark-up of changes if applicable
- i. Traceability (i.e. cut-in SN, date code)

In addition, a PCN shall be issued upon initial release to production (or pilot for HPR) and includes (or the links to) the required documents. It shall include the formal and controlled releases of specifications, drawings, manufacturing plans and others.

10.1. Burn-In (BI) and Ongoing Reliability Testing (ORT)

Burn-in test duration starts at a minimum of 24 hours. Unit-hours, acceptable failure rate, production yield requirements are defined by the customer before products are released to production.

No reduction is allowed during NPI. An approved PCN with the supporting documents (including but not limited to BI test history, closed RCCA) is required before implementation. Any manufacturing location change will reset test time and unit count to 24 hours.

Upon agreement, test time may be increased if needed and as applicable depending on the failure mechanism.

Unless otherwise agreed upon, the required test conditions are as follow:

Input Voltage: Nominal input voltage with cycling

Output Loading: 95 +/-5% of maximum rating with cycling

Current sharing: Yes

Ambient Temperature: 50°C +/- 5°C (operating temperature is only 45°C)

Monitoring: Automated monitoring of input and output voltage and FW status reporting

Ongoing Reliability Testing (ORT) shall be performed when the minimum burn-in time is reached.

Sampling size will be agreed between the customer and the supplier.

10.2. Hipot and Ground Continuity Test

The PSU shall be tested 100% in production for both Hi-Pot (with the applicable limits for the AC leakage current) and ground continuity (per the applicable standard). Stamps shall be applied to the chassis proving that both tests passed in production.

10.3. DFT and DFM

- The Vendor shall provide DFT and DFM reports at EVT and/or DVT phase.
- For all the boards used in the PSU design: the in-circuit test coverage and test point access shall be 95% or higher. The rest shall be covered by functional testing.

10.4. Quality Control Process

- All assembly inspection dimensions and tolerances in the drawings provided must be met.
- Incoming Quality: <0.1% rejections.
- CPK values should equal or exceed 1.33 (Pilot Build and Production) using a sample size equal to 32 for CPK measurement.
- CPK measurement should be performed on every batch/lot build of pilot and mass production. During production, weekly CPK reports must be sent to the customer.
- The Vendor will implement further quality control procedures during production, by sampling power supplies randomly from the production line and running full tests to prove ongoing compliance to the requirements (it may include EMI production). Process shall be documented and submitted to the customer.
- PCB boards are UL recognized components, rated 94 V-0 and 130°C MOT.
- Multi-layers (> 2 layers) PCB boards are welcome for a better layout and simplification of the manufacturing process as needed.
- The Vendor shall maintain a one to one system traceability up to component level.

10.5. Mass Production First Article Samples

Prior to final project release and mass production, the Vendor will submit to customer a good quantity of PVT production pilot run verification samples, including the following documentation:

- All the pertinent development docs, production docs, and reports necessary for the customer to release the product for mass production.
- The pilot samples shall be built in the allocated facility for mass production and use hardtooled chassis and parts (where applicable).

- Full spec compliance matrix, full test report, production line final test 'pass' tickets.
- Samples passed the burn in-process planned for production.
- Samples are shipped using the approved for production shipping box.
- The units are certified and with safety labels applied ("Pending Certification" sticker may be allowed until the certification process is complete).

10.6. Spec Compliance, Quality FA, Warranty

- The Vendor is responsible for the PSU to meet the specifications as stand-alone unit as well as at system level, and for assuring that the power supplies shipped in production will conform to the specifications with no deviations.
- A specification compliance matrix and test report shall be submitted to the customer for each PSU revision: EVT(P1), DVT(P2), and PVT (Pilot).
- The Vendor is responsible for exceeding production quality standards achieved on the pilot run built without fluctuations.
- All failures from EVT, DVT, PVT are required to complete failure and root cause analysis and report corrective action prior to entering mass production.
- Standard failure analysis turnaround time on RMA units shall be 7 days upon receipt at the Vendor's facility. For urgent units, FA may be requested to be provided within 24hrs upon receipt. An 8D report may be required on trending issues and/or those causing thermal events.
- The Vendor shall warrant the power supply for defects and workmanship for a period of three (3) years from the date of shipment when the device is operated within specifications. The warranty is fully transferable to any end user. A standard "VOID" warranty sticker may be applied.

10.7. Conformal Coating

The gate terminals of all high-voltage MOSFETs as well as the most sensitive areas of the electronic circuitry in the boards must be protected against potential resistive contacts with other voltage potentials (e.g. MOS Drain leads, etc.) due to whiskers, pollutant particles, dust, moisture (for example contaminant substances can become conductive in presence of moisture), and up to some local wet condensation occurrences.

Local automated, conformal coating shall be applied to critical areas of the boards to protect sensitive circuitries, using an atomized spray process (no dipping process is considered here). The thickness of the coating shall be ~50µm to ~150µm. The Vendor (mainly the ultimate manufacturing facility) shall demonstrate proven capability and extensive experience on automated conformal coating application and processes. The power modules main board shall be partially conformal coated (top & bottom); critical board areas shall be agreed upon, typically the ac high- voltage input section at shelf level, like the AC input, board-mount connector area.

11. Compliance requirements

The power supply unit shall be designed for compliance to allow worldwide deployment. Additionally, the manufacturer is fully responsible for:

- Ensuring the complete compliance of the PSU in the environment it is intended to function (as described by the Rack Spec)
- Maintaining and updating the PSU safety reports to current requirements and all new released requirements.

- All design and recertification costs required to update the power supply to meet the new requirements.
- Meeting EMC requirements
- Meeting Safety requirements

The manufacturer is responsible for obtaining the safety certifications specified below.

11.1. Safety Standards

The product is to be designed to comply with the latest edition, revision, and amendment of the following standards. The product shall be designed such that the following safety certifications are obtainable: UL 62368-1, IEC 62368-1 and EN 62368-1; hazard-based performance standard for Audio video, IT & Communication Technology Equipment

The manufacturer shall obtain the following safety certifications for the PSU as applicable. Only requirements that absolutely rely on or are affected by the system may be left to the system level evaluation [i.e. minimize Conditions of Acceptability]. Below are common requirements for North America and Europe. For other countries, different certifications may be required:

- UL or an equivalent NRTL for the US with follow-up service (e.g. UL or CSA).
- CB Certificate and test report issued by CSA, UL, VDE, TUV or DEMKO
- CE Marking for EU

11.1.1. Safety requirements

Following (but not limited) safety requirements are applicable:

- All Fans shall have the minimum certifications: UL and TUV or VDE.
- All current limiting devices shall have UL and TUV or VDE certifications and shall be suitable rated for the application where the device in its application complies with IEC/UL 62368-1.
- All printed wiring boards shall be rated UL94V-0 and be sourced from a UL approved printed wiring board manufacturer.
- All connectors shall be UL recognized and have a UL flame rating of UL94V-0.
- All wiring harnesses shall be sourced from a UL approved wiring harness manufacturer. SELV cable to be rated minimum 80V, 125C.
- Product safety label must be printed on UL approved label stock and printer ribbon. Alternatively, labels can be purchased from a UL approved label manufacturer.
- The product must be marked with the correct regulatory markings to support the certifications that are specified in this document.
- Any single fault condition applied to the safety critical component shall not cause any electrical or mechanical unsafe conditions (such as electric shock to user, arcing to ignite fire...)
- The end product (power shelf) shall meet the acoustic noise requirement specified in FBPB: 18-000207
- PCBA shall meet the clearance and creepage requirements defined in FBPB: 18-000207, IEC/UL 62368 standard, whichever the greater.

11.2. EMC Requirements

The power supply shall meet the following requirements in the latest edition of standards when operating under typical load conditions and with all ports fully loaded;

The Power supply integrated into the shelf is called the component power supply. Manufacturer shall provide the proof of compliance for the component power supply that are required for spare parts shipment. The component power supply shall not contribute any noncompliant conditions to the end-use product.

If at any time it is found that a supplier's component power supply causes the end-use product to fail emissions and/or immunity testing, the supplier will be instructed to investigate and resolve the problem. The power shelf shall have minimum 6dB margin from the Class A limit for the radiated and conducted emissions. Depending on the system manufacturer's design goals and business needs, more margin may be required when it is integrated into the final end system.

The following EMC Standards (the latest version) are applicable to the product.

- FCC /ICES-003
- CISPR 32/EN55032
- CISPR 35/EN55035 - Immunity
- EN61000-3-2 - Harmonics
- EN61000-3-3 - Voltage Flicker

Each individual basic standard for immunity test has the following minimum passing requirement. Higher level of passing criteria may be applied depending on the system manufacturer's design goals and business needs.

- EN61000-4-2 Electrostatic Discharge Immunity
 - Contact discharge: >5.6kV
 - Air discharge: >11.2kV
- EN61000-4-3 Radiated Immunity
 - > 3V/m
- EN61000-4-4 Electrical Fast Transient Immunity
 - AC Power Line: >1kV
 - Signal Line: >0.5kV
- EN61000-4-5 Surge
 - AC Power Line: >2kV (Line-to-line), >4kV (Line-to-earth)
 - Signal Port: >1kV
- EN61000-4-6 Immunity to Conducted Disturbances
 - DC Power Line: > 3Vrms
- EN61000-4-8 Power Frequency Magnetic Field Immunity, when applicable
 - > 1A/m
- EN61000-4-11 Voltage dip and sag

11.3. Environmental Compliance

The power shelf (including all components inside) shall comply with the following minimum environmental requirement and manufacturer shall provide full material disclosure, Declaration of Conformity and technical documentation to demonstrate compliance. The system manufacturer may have additional requirements depending on its design goals and business needs.

- RoHS Directive (2011/65/EU and 2015/863/EU); aims to reduce the environmental impact of EEE by restricting the use of certain substances during manufacture
- REACH Regulation (EC) No 1907/2006 including the restriction of PFAS substances; registration with the European Chemicals Agency (ECHA), evaluation, authorization and restriction of chemicals.
- Halogen Free: IEC 61249-2-21, Definition of Halogen Free, 900ppm for Br or Cl, or 1500ppm combined

- US SEC conflict mineral regulation to source mineral materials from socially responsible countries, if applicable
- Waste Electrical and Electronic Equipment (“WEEE”) Directive (2012/19/EU) if applicable; aims to reduce the environmental impact of EEE by restricting the use of certain substances during manufacture
- The vendor shall provide the declaration of compliance document stating that a product doesn't contain any substances regulated by EPA 40 CFR751. Refer to FBPN: 18-000142 for more detail environmental compliance requirements

11.4. Documentation

The manufacturer shall provide reproducible copies of all pertinent documentation relating to the following:

- Product Information
- Bill of Materials
- Schematics
- functional test report
- Final Compliance Approval
- NRTL certificate and report, Conditions of Acceptability and test report plus User documentation that explains safe installation and operating procedures.
- CB Certificate and report, including schematics
- Manufacturer's Declaration of Conformity to EN 62368-1, EN55032, EN55035 and ROHS
- FCC Part 15 Class A and CISPR32 Class A test data
- Declaration of Conformity to EN 61000-3-2 Class A and test report including waveforms and harmonic output levels.
- Other applicable certificates required by the system manufacturer.

12. Compliance with OCP Tenets

12.1. Openness

This Specification publishes the complete set of requirements needed for any qualified third party to design, build, and service a compliant 12kW PSU, including the electrical and thermal requirements, mechanical envelope, connector pinouts, register maps, and firmware upgrade and metering interfaces. No proprietary interface or tooling is required to implement or integrate the module.

12.2. Efficiency

The PSU is designed for high conversion efficiency across its operating range, with active power factor correction, reducing facility-level energy use and operating cost. The Narrow Range 48V architecture improves end-to-end power delivery efficiency relative to the prior generation and enables efficient downstream conversion. Delivering 12kW in a single module also improves power density and rack-level utilization.

12.3. Impact

The 12kW PSU is a core building block of the Open Rack V3 High Power Rack (HPR) ecosystem, supplying DC power to the power shelf that distributes power across the rack. By standardizing a high-power, high-efficiency PSU interface alongside the associated power shelf, PMM, and BBU contributions, including pulse-load management and AC-loss-to-BBU transition behavior - it enables a global supply chain to deliver high-power rack infrastructure to many customers across many regions, accelerating industry adoption of the next-generation high-power rack standard.

12.4. Scale

The PSU is a modular, hot-swappable unit deployed with a minimum of N+1 redundancy and designed to operate across many power shelves in parallel, enabling deployment from single-rack configurations to large hyperscale clusters.

12.5. Sustainability

The Specification defines mandatory post-consumer recycled (PCR) content targets, met in aggregate by weight for polymers and steel, together with supplier qualification, recycled-content certification, and a BOM-level PCR breakdown to verify them. The high conversion efficiency, greater than 97.5% peak, directly reduces energy consumption and waste heat over the product lifetime.